Exercise 1: Exam 27/06/2024

Assume that $y_1, \ldots, y_{200}$ are realizations of independent Gaussian random variables with variance equal to 1 and and mean $\beta_1 + \beta_2 \exp\{z_i\}$ for $i = 1, \ldots, 120$, and mean $\beta_1 + \beta_3 \exp\{z_i^2\}$ for $i = 121, \ldots, 200$; where the z_i are known constants and $(\beta_1, \beta_2, \beta_3)$ are unknown real parameters.

a) Formulate a Gaussian linear model for the variables $Y_1, \ldots, Y_{200}$. Are the model assumptions satisfied?

b) State the parameter space and sample space.

c) Express the model in matrix form: $\boldsymbol{Y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, explicitly stating how $\boldsymbol{Y}$, X, $\boldsymbol{\beta}$, and $\boldsymbol{\varepsilon}$ are defined and their dimensions. Write the distribution of $\boldsymbol{Y}$ and $\boldsymbol{\varepsilon}$.

d) Obtain the expression of the matrix $X^T X$ and the vector $X^T \boldsymbol{y}$; state how these elements should be used to obtain the maximum likelihood estimate $\hat{\boldsymbol{\beta}}$.

e) Write the distribution of the maximum likelihood estimator $\hat{\boldsymbol{B}}$.

f) Let $\boldsymbol{e} = \boldsymbol{y} - X\hat{\boldsymbol{\beta}}$ be the vector of the residuals. State which of the following identities are true and motivate the answer:

$$\sum_{i=1}^{200} e_i = 0 \qquad \sum_{i=1}^{200} e_i z_i = 0 \qquad \sum_{i=1}^{200} e_i z_i^2 = 0$$

$$\sum_{i=1}^{200} e_i \exp\{z_i\} = 0 \qquad \sum_{i=1}^{200} e_i \exp\{z_i^2\} = 0 \qquad \sum_{i=1}^{120} e_i \exp\{z_i\} = 0$$

(hint: read the indices in the sum!)

$y_1, \dots, y_{200}$ $\quad n = 200$

$Y_i \sim N(\beta_1 + \beta_2 e^{z_i}, 1)$ indep $i = 1, \dots, 120$

$Y_i \sim N(\beta_1 + \beta_3 e^{z_i^2}, 1)$ indep $i = 121, \dots, 200$

a) $Y_i = \beta_1 + \beta_2 x_{i2} + \beta_3 x_{i3} + \varepsilon_i \qquad \varepsilon_i \sim N(0,1)$ iid

$$x_{i2} = \begin{cases} e^{z_i} & \text{for } i = 1, \dots, 120 \\ 0 & \text{for } i = 121, \dots, 200 \end{cases}$$

$$x_{i3} = \begin{cases} 0 & \text{for } i = 1, \dots, 120 \\ e^{z_i^2} & \text{for } i = 121, \dots, 200 \end{cases}$$

b) $\Omega = \mathbb{R}^3$
$\mathcal{Y} = \mathbb{R}^{200}$

c) $\underline{Y} = X\underline{\beta} + \underline{\varepsilon}$

$\underline{Y} = [Y_1, Y_2, \dots, Y_{120}, Y_{121}, \dots, Y_{200}]^T$ 200-dim vector of r.v.

$\underline{Y} \sim N_{200}(X\underline{\beta}, I_{200})$

X $\quad n \times p = 200 \times 3$ matrix of known constants

$$X = [\underline{1} \quad \underline{x}_2 \quad \underline{x}_3] = \begin{bmatrix} 1 & e^{z_1} & 0 \\ 1 & e^{z_2} & 0 \\ \vdots & \vdots & \vdots \\ 1 & e^{z_{120}} & 0 \\ 1 & 0 & e^{z_{121}^2} \\ 1 & 0 & e^{z_{122}^2} \\ \vdots & \vdots & \vdots \\ 1 & 0 & \\ & & e^{z_{200}^2} \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & e^{-\cdots} \end{bmatrix}$$

$\underline{\beta} = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix}$ 3-dim vector of unknown parameters

$\underline{\varepsilon} = [\varepsilon_1 \ \varepsilon_2 \ \dots \ \varepsilon_{200}]^T$ 200-dim vector of r.v.

$$\underline{\varepsilon} \sim N_{200}(\underline{0}, I_{200})$$

d) $X^T X = \begin{bmatrix} \underline{1}^T \\ \underline{x}_2^T \\ \underline{x}_3^T \end{bmatrix} [\underline{1} \ \underline{x}_2 \ \underline{x}_3] = \begin{bmatrix} \underline{1}^T\underline{1} & \underline{1}^T\underline{x}_2 & \underline{1}^T\underline{x}_3 \\ \underline{x}_2^T\underline{1} & \underline{x}_2^T\underline{x}_2 & \underline{x}_2^T\underline{x}_3 \\ \underline{x}_3^T\underline{1} & \underline{x}_3^T\underline{x}_2 & \underline{x}_3^T\underline{x}_3 \end{bmatrix}$

$(3\times n)(n\times 3)$

3×3

$$= \begin{bmatrix} 200 & \sum_{i=1}^{120} e^{z_i} & \sum_{i=121}^{200} e^{z_i^2} \\ \sum_{i=1}^{120} e^{z_i} & \sum_{i=1}^{120} e^{2z_i} & 0 \\ \sum_{i=121}^{200} e^{z_i^2} & 0 & \sum_{i=121}^{200} e^{2z_i^2} \end{bmatrix}$$

$$X^T\underline{y} = \begin{bmatrix} \underline{1}^T \\ \underline{x}_2^T \\ \underline{x}_3^T \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_{200} \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^{200} y_i \\ \sum_{i=1}^{120} e^{z_i} \cdot y_i \\ \sum_{i=121}^{200} e^{z_i^2} \cdot y \end{bmatrix}$$

$(3\times n)(n\times 1)$

3×1

$$\hat{\underline{\beta}} = (X^T X)^{-1} X^T \underline{y}$$

e) distribution of $\hat{\underline{B}}$: $\hat{\underline{B}} \sim N_3(\underline{\beta}, (X^T X)^{-1})$

usually $(X^T X)^{-1}\sigma^2$ but here $\sigma^2 = 1$

f) we know $\underline{e}^T \underline{v} = 0$ for all $\underline{v} \in C(X)$ (column space of X)

i) $\sum_{i=1}^{200} e_i = 0 \iff \underline{e}^T \underline{1} = 0$ True $\underline{1}$ is the first column of X

ii) $\sum_{i=1}^{200} e_i z_i = 0$ no: $\underline{z} \notin C(X)$

iii) $\sum_{i=1}^{200} e_i z_i^2 = 0$ no: $\underline{z}^2 \notin C(X)$

iv) $\sum_{i=1}^{200} e_i e^{z_i} = 0$ no

v) $\sum_{i=1}^{200} e_i e^{z_i^2} = 0$ no

vi) $\sum_{i=1}^{120} e_i e^{z_i} = 0$ yes $\underline{e}^T \underline{x}_2 = 0$